

2022-Ph-H1-Q25

Section: Electricity

Topic: Capacitors

Which graph correctly shows how the voltage across a capacitor increases as it charges?

Answer: D

Guidance for Students:

The voltage increases **exponentially**, approaching the supply voltage but never quite reaching it. It's a **curve**, not straight.

Revision Tips:

- Charging graph: exponential rise
 - Time constant $\tau = RC$ affects shape
 - Know discharge is an exponential **decay**
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